Rui Liu

Artificial Intelligence Undergraduate

github.com/QLruil419

Profile

I am an undergraduate student in Artificial Intelligence at Xi'an Jiaotong-Liverpool University. My academic projects span database design, graph-based route planning, algorithms, and computer systems simulation. Through these projects, I have gained experience in organizing data, implementing algorithms, and analyzing system performance.

Education

Xi'an Jiaotong-Liverpool University
Major: Artificial Intelligence
August 2024 – July 2028 (expected)

Skills

Programming	Python, Java, SQL
Databases	MySQL, PostgreSQL
Graph Technologies	Neo4j, Cypher
Data Modeling	Entity–Relationship Modeling, Database Normalization, XML, DTD
Numerical Computing	NumPy
Algorithms	Divide and Conquer, Greedy Algorithms, Dynamic Programming, Shortest-Path Algorithms

Selected Projects

Data Modeling and Graph-Based Route Planning

Core Team Member | June 2026

Contributed to relational database design, PostgreSQL queries, and graph-based route planning using Neo4j and Cypher. Modeled hierarchical e-commerce order data using XML and DTD.

Algorithms and Data Structures

Core Team Member | May 2026

Applied divide-and-conquer, greedy, and dynamic programming methods to algorithmic problems. Worked with hash tables, heaps, Dijkstra's algorithm, and 0–1 BFS for order management, task scheduling, and shortest-path computation.

Computer Architecture and Systems Simulation

Core Team Member | April–May 2026

Contributed to concurrency control and simulations of cache replacement, disk scheduling, and CPU scheduling. Used NumPy vectorization and algebraic reformulation to improve numerical stability.